

Without a tedious menu tree to deal with 2M point memory waveform, GDS-1000A gives a user-friendly operation procedure to expand any part of the waveform and display it in detail on the screen. Just use the horizontal position knob to move the zoom range sideways, and time base knob to change the zoom range. Press 'window zoom' and the expanded waveform is there.

Yokogawa claims to be the only company in the world to provide 8-channel, 500MHz DSOs. Its WT200 series digital power meters measure low powers, making them suitable for sub-watt applications like solar, LED panel, automotive, robotics, fuel-cell and space technologies.

Significant trends

As the global economic climate places additional constraints on budgets, test engineers are challenged to identify ways to test devices more efficiently than ever before. National Instruments has identified three trends—software-defined instrumentation, parallel processing technologies and new methods for wireless and semiconductor test—that will significantly improve the efficiency of test and measurement systems in 2009. These trends help engineers develop faster and more flexible automated test systems while reducing their overall cost of test, and companies worldwide and from all industry segments are seeing significant benefits from applying these methods and technologies.

Software-defined instrumentation. Engineers are using software-defined instrumentation to achieve new levels of measurement performance and lower test costs by applying the latest technological advancements such as multicore processing and FPGAs in their test systems to meet the demands of new application areas such as wireless and protocol-aware test. The quick return on investment from these benefits is contributing significantly to the mainstream adoption of software-defined instrumentation.

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By making test and measurement devices software based one can do a lot more with them than otherwise. Software adds innumerable advantages such as flexibility, customisability, scalability, re-use, ease of use, low maintenance, faster development, reduced time-to-market, and all this contributes to reducing engineering costs in many different ways. More and more features and capabilities are being added by upgrading the device software rather than the hardware. This helps in keeping the costs down.

Consistent hardware and software concept. The reuse of the tester hardware, tester software and test routines in all phases of the development process—from the first software module test to the conformance tests and regression tests, as well as for optimising performance, delivers optimum results.

Better test coverage through automation. Most tests can be fully automated if necessary. This approach allows users to perform a large number of tests for the various versions of the mobile phone software. This increases test throughput.

Multitechnology solution. With mobile phones supporting several technologies on a single device, it is desirable that the testing equipment also support multiple technologies on a single box.

The future

For most of the general-purpose measurements the old instruments work just fine. But for testing new applications, which place a stringent requirement on such parameters as power, bandwidth and speed, the design engineers will need to upgrade to new instruments.

A migration is expected from older T&M technologies to newer and more advanced technologies. One example is LXI or LAN extension for instrumentation slowly replacing GPIB or general-purpose instrumentation bus. There is a natural application space for new instruments, e.g., semiconductor technologies, nano and organic materials, solar cells and MEMS to name a few.

Companies will move towards availing T&M services rather than purchase of T&M. Most of the T&M manufacturers are not geared up to meet this requirement as this would require a major shift in their policy of approach to the market.

If we talk about sales and marketing of T&M instruments, the concept of mall-like experience is being introduced by T&M providers like Glorious Electronics. Unlike the typical 'instruments-selling shops' where the client is supposed to have already made a choice, or is purely guided by the 'lowest cost' syndrome without understanding the application or need, here he is guided and provided with a demonstration if necessary. ●

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